

Do you want to dive deep into neural networks? Does the sheer amount of complexity and scientific knowledge of neural networks seem too overwhelming to understand? Neural networks are quickly becoming an important part of modern machine learning systems that require a certain knack to work on. They have seeped into several domains of the industry and continue to be a subject of wide interest among researchers and futurists.

What is a neural network

Like a biological neuron that routes out to synapses and other neurons, a neural network closely mimics a machine schematic where each node represents an input. These inputs are well stacked over layers and cross over each other with some initial values(parameters and biases).

In these sequence of the nodes, the machine 'learns' or adapts itself to the input and compares the output produced to the actual results. The fundamental theory behind neural networks works on the principle that inputs, when tested with different weights(coefficients) over several iterations, can ultimately help the system learn how to arrive at the output.

All the values that cross each other meet at other junction nodes from where they intersect once again to the final output. This is where the fun part comes in.

The analyst has to select the number of iterations, the training algorithm used and the number of hidden layers to arrive at the output. Neural networks thus become a schematic runway for data to conform to the desired result. The more the inputs and outputs, the more complicated and ingenious the network have to become to get good accuracies.

From a mathematical standpoint, the nodes themselves add to a formulaic scheme of the network. The nodes are the inputs which when multiplied with their weights and then added to the bias produces the output.

Some of the typical components of any neural network program include:-

Input Layer: This refers to the initial layer where all the inputs of the problem are inserted and fit into an appropriate number of nodes.

Activation Function: Since input values are rarely in the same unit, an activation function is added to normalise all the values to a single base order which is then inserted along with the inputs at the input layer.

Activation functions themselves are elaborate equations with a predefined algorithm used to normalise data and can differ depending on the problem statements.